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Improved gap filling approach and uncertainty estimation for eddy covariance N_2O fluxes

J.P. Goodrich [a,*], A.M. Wall [a], D.I. Campbell [a], D. Fletcher [b], A.R. Wecking [a], L.A. Schipper [a]

[a] University of Waikato, School of Science, Hillcrest, Hamilton, New Zealand
[b] University of Waikato, School of Computing and Mathematical Sciences, Hillcrest, Hamilton, New Zealand

A B S T R A C T

Agricultural nitrous oxide (N_2O) emissions comprise a majority of the global source of this powerful greenhouse gas. Mitigation approaches for reducing emissions are difficult to evaluate at appropriate field scales because of the substantial effort and expense associated with relatively new technology allowing eddy covariance measurements of N_2O fluxes (F_{N2O}). Here we present a new approach for gap filling eddy covariance F_{N2O} and estimating annual uncertainties for a temperate grazed grassland. We tested the potential of using one flux tower to evaluate emissions mitigation options in one paddock relative to an adjacent, unchanged paddock by partitioning data by source footprint contribution. Because of the complexity of spatiotemporal controls on F_{N2O}, we generated a large set of environmental variables and features as input for machine learning algorithms. Inputs were transformed using partial least squares (PLS) decomposition, isolating features with the greatest influence on F_{N2O}. PLS scores were fed to both a neural network (NN) and a locally-weighted k-nearest neighbours (kNN) regression. While the NN and kNN preformed similarly well, kNN regression accounted for the largest proportion of variance (52-72%) and resulted in the lowest bias for each of the three source footprint areas (full footprint and two separated adjacent paddocks, P53 and P54). Annual uncertainty estimates included random measurement uncertainty, accuracy and precision of the gap filling approach, and uncertainty associated with choice of threshold for atmospheric turbulence filtering and footprint contributions. Total N_2O emissions for the full footprint, P53, and P54 were 7.4 $\pm$0.35, 7.7 $\pm$0.80, and 6.4 $\pm$0.63 kg N_2O-N ha^{-1}, respectively in Year 1, and 6.9 $\pm$0.33, 7.3 $\pm$0.63, and 6.7 $\pm$0.63 kg N_2O-N ha^{-1}, respectively in Year 2. These 95% confidence intervals on the annual F_{N2O} suggest that we could detect differences of 10-15% between paddocks at this site when testing mitigation options.

1. Introduction

Nitrous oxide (N_2O) is a powerful, long-lived greenhouse gas (IPCC, 2013), and agriculture is the largest source globally (Smith et al., 2008a). Both nitrification and incomplete denitrification of excess nitrogen in soils produce N_2O depending on soil physical characteristics, carbon content, moisture content, and mineral nitrogen availability (Smith, 2017). Grazed pasture is of particular concern because of the concentrated nitrogen loads deposited in urine patches by grazing animals (e.g., de Klein et al., 2003), which result in hotspot emissions of N_2O, especially following rain events (Liang et al., 2018).

In New Zealand, managed grasslands cover 26% of the terrestrial surface (Suttie et al., 2005), and the national greenhouse gas budget reflects a heavily agricultural economy, with CH_4 and N_2O making up 53% of total emissions (MfE, 2020), in contrast to most developed countries, where CO_2 emissions dominate. This has led to national efforts aimed at developing management strategies with demonstrated potential for reducing agricultural greenhouse gas emissions without compromising economic output. To date, much of the research into N_2O emissions from grazed pasture, in New Zealand and abroad, has been based on static chamber trials (e.g., Rochette 2011; Rochette et al., 2008; Lammirato et al., 2018) to test mitigation options or to impose treatments and estimate emission factors (Luo et al., 2008; Luo et al., 2007). While providing a pragmatic approach for constructing regional or national scale budgets, chamber-based information does not fully capture the heterogeneous spatiotemporal variability characteristic of N_2O fluxes (F_{N2O}) in grazed systems. For example, Wecking et al. (2020), demonstrated that a standard chamber-based emission factor approach does not sufficiently account for consistent background fluxes throughout the year, a clear drawback leading to under-estimates of annual totals. Furthermore, pulse emissions of N_2O following rain events occurring relatively soon after grazing are highly variable in magnitude and duration (e.g., Liang et al., 2018). In general, the information required to mechanistically understand these pulse emissions does not currently exist at field or paddock scales, and therefore we cannot reliably scale plot-level information across space and over time. This makes it difficult to determine if and to what extent any given mitigation option reduces emissions at relevant field scales (Wecking et al., 2020; Cowan et al., 2020).

The relatively recent commercial availability of high-frequency N_2O

* Corresponding author.
E-mail address: jordan.goodrich@waikato.ac.nz (J.P. Goodrich).

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analysers has allowed for eddy covariance F_{N2O} measurements at paddock scales in a growing number of locations around the world (Merbold et al., 2014; Rannik et al., 2015; Kroon et al., 2010; Cowan et al., 2016; Hörtnagl and Wohlfahrt, 2014), including New Zealand (Liang et al., 2018). While offering new insights regarding spatiotemporal controls on agricultural F_{N2O}, the expense of eddy covariance N_2O analysers usually precludes the deployment of multiple towers to test emissions mitigation strategies. However, recent studies have begun to test approaches for comparing cumulative fluxes from adjacent parcels of land using one flux tower. Fuchs et al. (2018) analysed F_{N2O} from a managed grassland that was mown and fertilized (i.e. not grazed), with the aim of evaluating effects of a change in management in one parcel while maintaining an adjacent parcel unchanged. Wall et al. (2020), also demonstrated this approach for CO_2 flux (F_{CO2}) in a grazed dairy pasture using one tower situated on a fence-line by partitioning data using footprint contribution thresholds. The result is effectively two separate flux time series. One drawback of this approach is that the data coverage of the split datasets is reduced by more than half of the original full footprint time series, where the extent of this reduction depends on contribution thresholds and predominant wind characteristics. For F_{CO2}, this problem can be handled using fairly standardized approaches to gap-filling and uncertainty estimation (Moffat et al., 2007; Reichstein et al., 2005; Elbers et al., 2011; Dragoni et al., 2007), aided by a much better understanding of what drives spatiotemporal variability in CO_2 exchange than F_{N2O} at field scales. There is yet no standard approach for filling gaps in F_{N2O} (Nemitz et al., 2018), and uncertainties associated with temporally integrated fluxes are rarely estimated. Fuchs et al. (2018) calculated daily averages from 10-min flux observations to compare F_{N2O} on days when measurements were available from both parcels, and estimated mean annual F_{N2O} by bootstrapping the available observations. While this approach does allow for estimating 95% confidence intervals on the distribution of resulting annual numbers, other sources of uncertainty may also be important to consider when comparing fluxes from adjacent parcels.

Liang et al. (2018) published the first year of data from the Troughton Farm site covered in our our analysis by adopting a two-fold approach for filling gaps in F_{N2O}. Missing values were linearly interpolated, with the exception of one large winter gap, filled using a multiple regression driven by soil temperature and water-filled pore space (WFPS). This approach would be very limited in situations with more frequent long gaps or more intermittent coverage, such as when partitioning the fluxes to adjacent paddocks.

Several other approaches for filling gaps in micrometeorological F_{N2O} have been utilized. Taki et al. (2018) compared linear interpolation with a neural network approach to gap filling flux gradient measurements of N_2O over cropland. They showed that the neural network captured a larger proportion of variance and resulted in lower annual bias, particularly when gap sizes increased. Cowan et al. (2020) used a general additive model (GAM) to fill gaps in F_{N2O} over an intensively managed grazed grassland. The GAM approach accounts for non-linear effects using combinations of smoothed splines that can be tuned to accommodate complex interactions among different drivers of F_{N2O}, and also allows for simulating many runs to estimate uncertainty (Cowan et al., 2020). However, in providing recommendations to the Integrated Carbon Observation System (ICOS) regarding F_{N2O}, Nemitz et al. (2018) could not suggest a universal approach to gap filling because of the mechanistic complexity and diversity of ecosystems. The authors suggested a pragmatic solution of combining linear interpolation of half-hourly measurements to estimate daily integrals for days with at least ~60% of available observations and remaining daily gaps can be filled with a moving average. There is clearly no consensus within the community on an ideal gap filling approach, in part because of the array of ecosystems and agricultural settings in which F_{N2O} measurements are being made, and the associated heterogeneity of controlling factors that may drive variability in the fluxes. A similar issue faces eddy covariance methane flux gap filling, for which there are not universal models that perform well across different sites, leading more groups to explore machine learning approaches for solutions (e.g., Dengel et al., 2013; Goodrich et al., 2015; Kim et al., 2020).

Our primary objective with this analysis was to develop a robust gap filling strategy for eddy covariance F_{N2O} that can be used to estimate seasonal and annual totals with associated uncertainties. We tested the skill of this approach in determining annual sums from two adjacent grazed paddocks after partitioning data based on footprint contribution. Operationally, our aim was to determine the minimum detectable difference, on an annual scale, between the two paddocks using one flux tower situated at their boundary. While the focus of this analysis was not on the comparison of different management strategies, our intent was to determine the feasibility of doing so with respect to F_{N2O} at this site. If sufficiently low detection of difference were possible, this approach could allow testing of paddock to farm-scale mitigation strategies to reduce F_{N2O} and also be coupled to CO_2 and CH_4 for full greenhouse gas budgets.

2. Methods

2.1. Site description

Troughton Farm is a commercially operating rotationally grazed dairy farm with a stocking rate of ~2.5 cows ha^{-1} located in Waharoa, New Zealand (37.78 °S, 175.80 °E). The farm consists of 67 paddocks roughly 3 ha in size that are rotationally grazed year-round. Each paddock is grazed once every 3—8 weeks, with greatest frequency during spring and lowest frequency during winter. Long-term mean annual temperature and precipitation determined from a nearby climate station were 13.3°C and 1250 mm (NIWA, 2018). More comprehensive description of the site and farm management were included in previous studies (Rutledge et al., 2017a; Rutledge et al., 2017b; Liang et al., 2018; Wecking et al., 2020). The paddock fences were re-aligned on 5 March 2018 in order to optimize the location of the flux tower at the fence boundary and test changes in management using a single tower, while working within the practical needs of the farmer (Fig. 1). In conjunction with the fence realignment, one of the paddocks (P54) was renewed by spraying off with herbicide and direct drilling with a mix of plantain, ryegrass, red and white clover, while the other (P53) remained under traditional ryegrass and clover mix. We used the realignment as a benchmark from which to evaluate annual F_{N2O} over two years. Year 1 includes data from 6 March 2017 to 5 March 2018, and year 2 includes data from 6 March 2018 to 5 March 2019. As such, the P53 and P54 footprints from year 1 actually cover portions of two different paddocks (separated by the dotted line in Fig. 1). Despite slight differences in the timing of grazing, the farm-scale management was identical for these paddocks over the course of year 1. Note that the degree of overlap in footprint contours was largely determined by the contribution thresholds we used.

2.2. Eddy covariance observations and flux processing

We used a closed path continuous-wave quantum cascade laser absorption spectrometer (CW-QCLAS; Aerodyne Research Inc., Billerica, MA, USA) for measurements of N_2O, and a CSAT3B sonic anemometer (Campbell Scientific Instruments, Logan, UT, USA) for observations of three-dimensional wind speed. The anemometer and gas inlet were mounted 2 m above ground. Sample air was drawn through a 6.1 m heated sample tube to the analyser using a dry scroll vacuum pump (XDS35i, Edwards, West Sussex, US) at 15 L min^{-1}. The analyser was housed within a temperature-controlled (set point ± 0.2°C) enclosure, which eliminated diel temperature variations. More details regarding the flux calculations and corrections are included in Liang et al. (2018) and Wecking et al. (2020). Briefly, high frequency data were processed to 30-minute fluxes using EddyPro 6.2.2 (LI-COR Inc., Natick, MA, USA). We used the Moncrief et al. (1997) fully analytic approach for high-pass

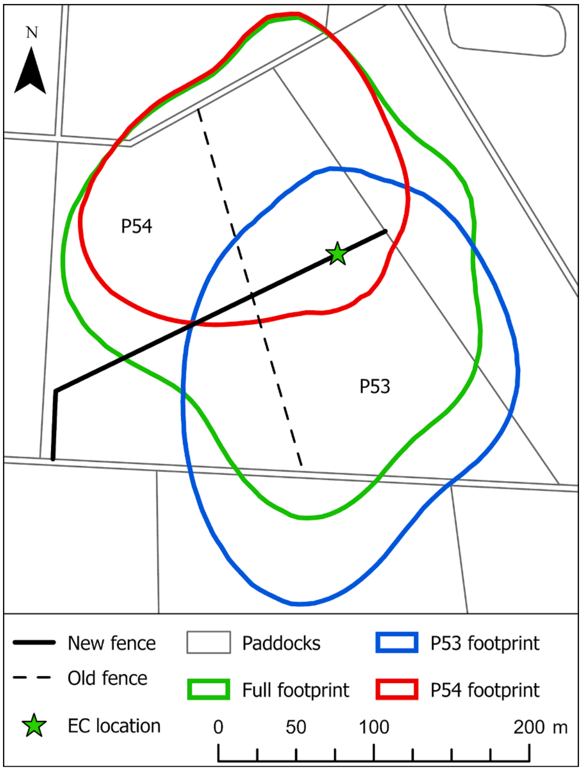

Fig. 1. Study site map showing the location of the eddy covariance tower at the boundary of P53 and P54. The footprint boundaries indicate the 80% source area contour for the three target source areas.

filtering, and the Fratini et al. (2012) *in situ* approach for low-pass filtering. Half-hourly F_{N2O} were filtered for out-of-range values (less than -1 nmol m^{-2} s^{-1} or greater than 20 nmol m^{-2} s^{-1}) and we used the 0-1-2 quality flag system of Mauder and Foken (2006), rejecting F_{N2O} flagged as 1 or 2. We further rejected half-hourly F_{N2O} associated with poorly developed atmospheric turbulence using a threshold of 0.1 m s^{-1} for the standard deviation of vertical wind speed (Wecking et al., 2020). Finally, periods during equipment maintenance and when cows were present were discarded from the analysis.

Footprint information was calculated based on Kormann and Meixner (2001), and we imposed footprint contribution thresholds to ensure that fluxes originated from the areas of interest. During daytime half-hours (photosynthetic photon flux density, PPFD >= 20 μmol m^{-2} s^{-1}), a 70% contribution was required, while at night (PPFD < 20 μmol m^{-2} s^{-1}), 60% contribution was required. This was done to accommodate generally lower nighttime wind speed and subsequent expansion of footprint source area. These thresholds were applied to paddock 53 (P53) and paddock 54 (P54), while the full footprint here was considered as P53 + P54 for consistency (Fig. 1). The data coverage for the full footprint was 49% in Year 1, and 45% in Year 2. For P53, data coverage was 10% in Year 1 and 9% in Year 2, while for P54 coverage was 15% in Year 1 and 17% in Year 2. The distributions of gaps in the P53 and P54 time series were similar to each other, and despite an increase in multi-day gaps after partitioning data by footprint contribution, a large majority of gaps were less than 1 day in length (Fig. 2). The largest gaps were caused by two ~2-week periods (in July 2017 and December 2018) when the QCLAS required maintenance and affected all data sets (Figure S1). Reduced turbulence at night resulted in lower data coverage as more data were rejected by the turbulence filtering step than during daytime as has been reported previously at other locations on this farm (e.g., Wall et al., 2020).

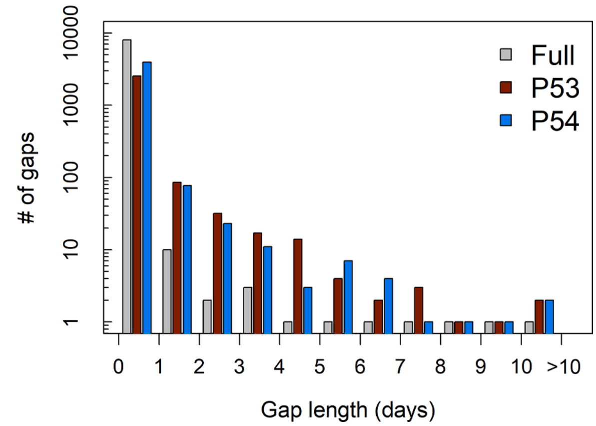

Fig. 2. Histogram of gap length (log scale) for the full footprint data set (Full), paddock 53 (P53), and paddock 54 (P54) from 6 March 2017 to 5 March 2019.

2.3. Preparation of F_{N2O} gap filling inputs

Volumetric soil moisture content (CS616; CSI) and soil temperature (107 type probes; CSI) were measured at 0.05 m, 0.10 m and 0.20 m depth. Rainfall was measured using a tipping bucket rain gauge (TB5; Hydrological Services). Net radiation and its components were measured with a four-component net radiometer (NR01, Hukseflux Thermal Sensors, Delft, Netherlands), PPFD was measured with an LI200SZ quantum sensor (LI-COR Inc.), and air temperature and relative humidity were measured at 1.55 m using a HMP155 (Vaisala Inc., Helsinki, Finland).

The suite of meteorological and ancillary data was used both as input and to generate additional features containing information potentially useful to reproducing missing F_{N2O} (Figure S2). These gap-filling model inputs included CO_2 flux components (net ecosystem exchange (NEE), ecosystem respiration (ER), and gross primary production (GPP)), gap-filled and partitioned using a neural network approach identical to that described in Wall et al. (2020), and measured with an LI7200 (en) closed-path infrared gas analyser (LI-COR Inc.).

As additional input features (Table S1), we calculated days since grazing, total rain since grazing, and the number half hours since grazing with rain > 0.2 mm, based on grazing records and precipitation data. Grazing events represent acute nitrogen inputs through urine patches given that cows excrete 70-90% of their ingested nitrogen (Jarvis et al., 1995). From half-hourly precipitation, volumetric soil moisture content (VMC), soil temperature, PPFD, wind speed, vapour pressure deficit (VPD), ER, GPP, and NEE data, we calculated running means or sums ranging from 1 hour to 80 days. Additionally, from ER and precipitation data, we also computed the maximum half-hourly values over the same time windows. Using this set of half-hourly and cumulative data, we calculated all possible cross-products and retained any resulting in Pearson's correlation coefficient (ρ) with F_{N2O} > 0.4. All data were standardized by subtracting the respective mean and dividing by the standard deviation.

To align training and validation of the gap filling approach and minimize the potential for introducing bias, we generated gaps in the filtered F_{N2O} data stratified by deciles of observed F_{N2O}. We held out 15% of the data for validation purposes during each run of the gap-filling algorithm, and by distributing this validation set evenly among deciles, we maintained distribution characteristics with the training sets.

The final step in preparing inputs for the gap-filling algorithm was a partial least squares (PLS) decomposition (). PLS is derived from singular value decomposition, which generates orthogonal principal components. However PLS differs by performing a dual decomposition of the input and target matrices, such that columns of resulting scores are ordered by the amount of target variance for which they account (Krishnan et al., 2011; Wold et al., 1987). Here the target matrix used in the PLS decomposition is the F_{N2O} vector. The number of PLS columns used as input for the subsequent gap filling approaches was then selected by minimizing the cross-validation error term of the associated PLS regression (PLSR). Dimensionality reduction is a common step referred to as 'feature selection' in machine learning, where PLS effectively solves the problem of multicollinearity while allowing the target variable (i.e. F_{N2O}) to inform that information which is retained as input for the gap filling algorithm (e.g., Hao et al., 2016; Yu et al., 2017). The input data sets generated before the PLS step for the full footprint, paddock 53, and paddock 54 described above had 303, 596, and 1081 columns, respectively, including the meteorological and soil data, grazing features, cumulative means, sums, and maximums, as well as the retained cross products. The PLS decomposition step reduced those to 120, 60, and 165 columns for the full footprint, paddock 53, and paddock 54, respectively, averaged over the 50 runs we performed (see below).

2.4. Gap filling approaches

We compared several approaches to filling gaps in F_{N2O}. The PLS regression itself can be used to estimate missing F_{N2O}, given the non-transformed gap-filled input data. Additionally, we used k-nearest neighbours (kNN) locally weighted regression, and an artificial neural network (NN). The kNN regression generates predictions from weighted averages of the k-nearest neighbours, where closer neighbours have larger weights. We used k=3 after testing the sensitivity of predictions to this parameter using metrics for variance and bias described below. The nearest neighbours were calculated using Euclidean distance in input space from the training data. 'Neighbours' here does not refer to adjacent measurements in time, but rather to observations where the combinations of input variables (i.e. PLS scores) were most similar. As such, larger values of k produce smoother predictions (lower variance, higher bias), and small values of k produce more spikey results (higher variance, lower bias). Weights were defined using a Gaussian kernel, resulting in larger weights for neighbours with smaller distances. We also used a neural network with three hidden layers, each with four nodes and hyperbolic tangent activation functions, a squared loss function, and stochastic gradient descent optimization.

Both algorithms (kNN and NN) were fed the reduced set of PLS scores as input. For each approach, we evaluated the performance based on R^2 of observed and predicted F_{N2O} from the validation set held back from the training phase. We also calculated the mean bias from residuals (predicted – observed) binned by observed F_{N2O} percentile within the validation set. Because our primary objective was to develop a robust gap filling strategy, we focused on the balance between maximizing variance explained and minimizing bias in the validation data sets rather than on the interpretability of the resulting models. We trained each approach 50 times using different training sets for each run, which was sufficient to obtain robust (i.e. unchanging) median values.

We also compared reference gap filling approaches to determine if the added complexity of machine learning algorithms provided meaningful improvements over previously reported approaches. Thus we filled missing F_{N2O} with 30-day running median of available observed data, and with linear interpolation of half-hourly F_{N2O}. To be useful when applying a gap filling approach to flux data partitioned into two different footprints, we needed to achieve reasonable predictions for long gaps (> 1 day). This is a particular concern for F_{N2O} considering the episodic nature of emissions and the potential for missing a pulse emission (Figure S3). As a way of testing this, we also held out data from a pulse emission period that occurred over four days (7 Mar 2018 – 11 Mar 2018) to evaluate whether the gap-filling approach could resolve a large episodic flux about which it had no information. We held back this pulse emission while training the gap filling model, and examined how well the predictions matched observations.

2.5. Uncertainty estimation

Uncertainties are rarely reported for F_{N2O}, and when they are, these generally apply to single components related to gap filling (e.g. Fuchs et al., 2018; Cowan et al., 2020). We have adapted approaches generally used for F_{CO2} in an effort to account for as many sources as pragmatically possible to provide confidence given the ultimate intention to compare annual numbers. We estimated random uncertainty in the filtered F_{N2O} measurements (σ_r) using an approach similar to Hollinger and Richardson (2005) paired-days technique. However, given concern over the episodic nature of F_{N2O}, depending especially on grazing frequency and rainfall, the paired-days approach may have resulted in artificially large uncertainty estimates, and we therefore used differences between adjacent half-hourly F_{N2O} (ΔF_{N2O}). We constrained the pairs by limiting the difference in footprint contribution, WFPS, and T_{soil} to < 5%. This approach assumes that fluxes calculated during two adjacent half-hours under similar conditions minimises any effects of management, and therefore any difference reflects random uncertainty in the eddy covariance measurements. In order to assign random uncertainty to every measured half hourly F_{N2O}, we fitted double exponential distributions to ΔF_{N2O}, binned by the mean of the pair, and used the resulting linear regression slope and intercept (Figure S4; Dragoni et al., 2007). Using the unique uncertainty distribution parameters for each measured half hour, we sampled from the distributions 1000 times, added these to the measured values and re-calculated annual totals (without adding uncertainty to gap-filled values). The final σ_r was calculated as the standard deviation of these 1000 annual totals. We use 1000 samples here since processing time was not an issue, in contrast to the model training step described above, where 50 runs were performed.

Uncertainty associated with the gap filling approach included two components. First, the accuracy was assessed in a similar way to the random uncertainty estimation described above using regression of binned uncertainties derived from kNN validation residuals. We again sampled 1000 times from associated distributions and applied them to gap filled values (σ_{GF1}) (Figure S5), this time without adding uncertainty to the measured half hours. The second component assessed the precision of the gap filling approach by calculating annual F_{N2O} for each of the 50 gap-filling runs and calculating the standard deviation of the resulting sample (σ_{GF2}).

Finally, we evaluated the uncertainty associated with the choice of thresholds for both atmospheric turbulence (σ_{turb}), represented by the standard deviation of vertical wind speed (σ_w), and for footprint contribution (σ_{FP}). For σ_w, we changed the default threshold (0.1 m s^{-1}) by ± 0.02 and 0.04 m s^{-1}, and for the footprint contribution, the default thresholds (70% during day, and 60% during night) were changed by ± 5% and ± 10% (Wall et al., 2020). Available data was reduced by about 5% as the σ_w threshold was increased from 0.06 to 0.14 m s^{-1}, and was reduced by 15% when increasing from minimum to maximum footprint contribution thresholds. The full gap filling approach was performed for each resulting dataset and the uncertainty associated with threshold choice was calculated as the standard deviation of the five annual F_{N2O} estimates (Wall et al., 2020).

Finally, overall uncertainty (σ_{FN2O}) associated with an annual sum was calculated as:

$$\sigma_{F_{N2O}} = \sqrt{\sigma_r^2 + \sigma_{GF1}^2 + \sigma_{GF2}^2 + \sigma_{turb}^2 + \sigma_{FP}^2} \tag{1}$$

By combining the uncertainty estimates using Eq. (1), we assume each term is independent. This is a conservative approach as we acknowledge, for example, that measurement uncertainty can be subtracted from gap-filling uncertainty to remove the redundant contribution (Dragoni et al., 2007).

All data post-processing, analysis, and gap-filling was done using Matlab (MATLAB R2019a, Mathworks Inc.) and R (R Core Team, 2018). The program code to prepare the input variables and perform the gap filling in R is available from the corresponding author upon request.

3. Results

3.1. Evaluating gap filling approaches

The kNN algorithm best accounted for variance in the observed validation F_{N2O} (52-72%), although the NN performed very similarly (48-68%), while PLSR resulted in the lowest R^2 (24-27%) for the full footprint, P53, and P54 datasets across both measurement years (Fig. 3). The gap filling bias followed a similar pattern among algorithms, characterized by zero or slightly positive bias (overestimation) at low F_{N2O}, and negative bias at high F_{N2O}, representing an underestimation of pulse emissions (Fig. 4). The kNN algorithm again consistently resulted in the lowest bias of the three techniques, particularly at the high end of the distribution, suggesting this algorithm is best suited for filling gaps in F_{N2O} at this site. We note that the 30-day median and the interpolation-based gap-filling approaches could not be evaluated with these same criteria because there is no unambiguous way to generate training and test sets, or some analogue, from which to calculate skill metrics, or uncertainties.

The test pulse F_{N2O}, held back from training the kNN algorithm, lasted approximately four days from 7 Mar 2018 to 11 March 2018 (Fig. 5). Fig. 5a illustrates the pulse period gap-filled using kNN, 30-day medians, and linear interpolation allowing each approach to access the available observations when making predictions. The 30-day median F_{N2O} reflects a highly skewed distribution in which such pulse emissions punctuate an otherwise low-magnitude baseline. With relatively short gaps spread throughout the pulse emission period, both linear interpolation and kNN generated F_{N2O} that fell within the observed pattern. Fig. 5b shows the gap-filled result for this test pulse when treating the full four-day period as a gap. Both the 30-day median and linear interpolation approaches are unsuitable for this situation because they do not generate any variability or come close to the observed F_{N2O} magnitudes. While somewhat under-estimating magnitudes (Fig. 5b), the kNN approach to filling such a multi-day gap containing a large pulse emission did reproduce the general behaviour of F_{N2O}. There were approximately 30 periods over the two-year study identified as pulse emissions using the full footprint dataset.

3.2. Annual F_{N2O} and uncertainties

We used the kNN gap-filled F_{N2O} to estimate annual totals from the full footprint, P53, and P54 by calculating the median half-hourly value generated from 50 kNN gap filling runs. Each of the uncertainty components were combined in quadrature resulting in a single σ_{FN2O} value for each year for each source area (Table 1). Subtle differences in the estimated uncertainty components can be attributed to intuitive features of the approach we have taken. For example, σ_r was larger for the full footprint data set than for the partitioned P53 and P54 data, which likely represents contributions from a larger area covering more heterogeneity across paddocks, whereas fluxes separated by paddocks are likely to be more similar. Also, σ_{GF2} was largest for P53, which had the largest proportion of gaps, leading to lower precision in the gap filling

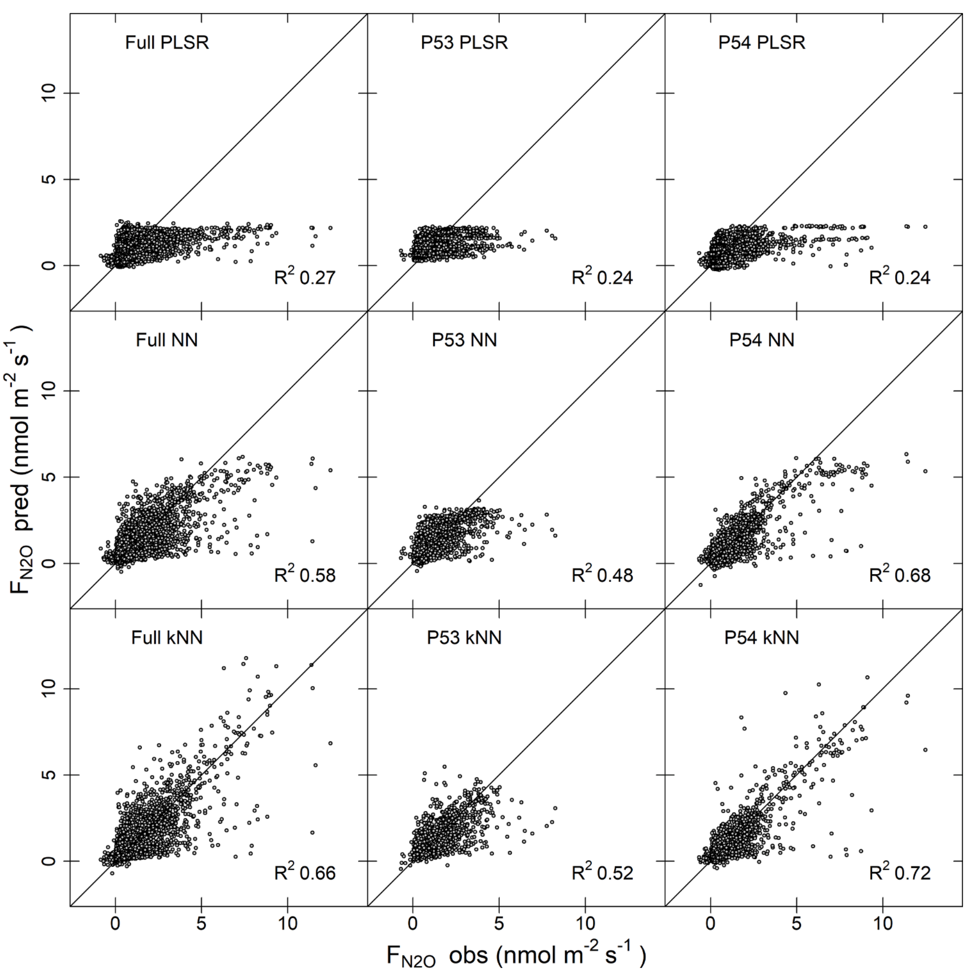

Fig. 3. Scatter plots showing observed (obs) and predicted (pred) F_{N2O} from the validation sets used to evaluate the three algorithms — partial least squares regression (PLSR), neural network (NN), and k-nearest neighbours regression (kNN), and for the three source areas – Full footprint, P53, and P54. The 1:1 lines is shown in black and R^2 values indicate variance explained.

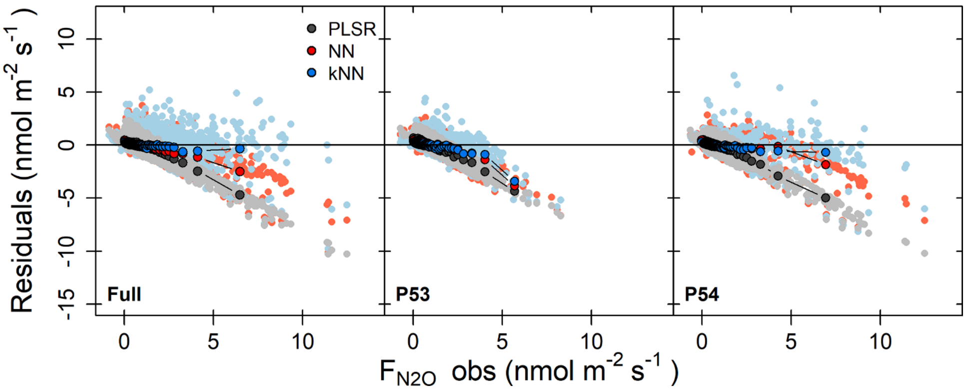

Fig. 4. Gap filling residuals from the validation data sets for the three algorithms and source areas as a function of observed F_{N2O}. Darker symbols are median residuals binned by F_{N2O} percentiles.

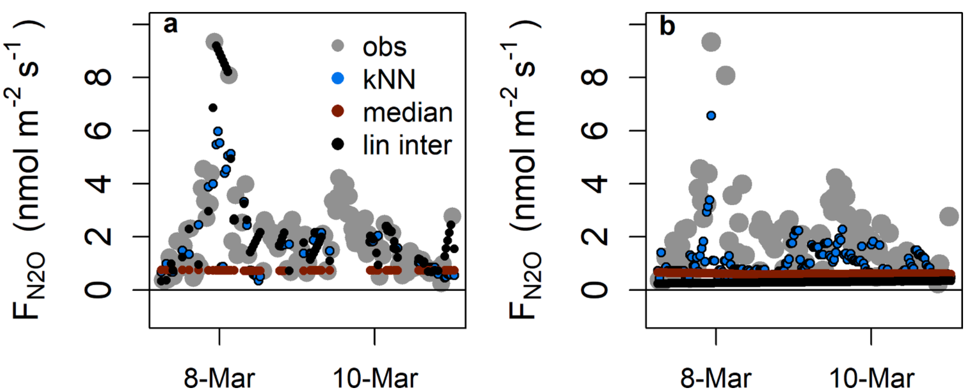

Fig. 5. A portion, 7 March 2018 – 11 March 2018, of the full study period showing observed F_{N2O}, as well at that gap-filled by kNN, 30-day running medians of the available half-hourly observations, and linear interpolation (a). Panel b shows the gap filling results when this pulse emission period is completely withheld from each algorithm.

Table 1
Uncertainty components in units of kg N_2O-N ha^{-1} for each of the source areas of interest (full footprint, P53, and P54). Subscript meanings *r:* random measurement uncertainty, *GF1*: random gap filling uncertainty, *GF2*: annual gap filling precision, *turb*: uncertainty associated with turbulence filter threshold choice, *FP*: uncertainty associated with footprint contribution threshold choice. σ_{FN2O} is the combined uncertainty according to Equation 1.

		Full	P53	P54
σ_r	Year 1	0.03	0.01	0.01
	Year 2	0.03	0.01	0.02
σ_{GF1}	Year 1	0.03	0.04	0.03
	Year 2	0.03	0.03	0.03
σ_{GF2}	Year 1	0.11	0.31	0.21
	Year 2	0.14	0.27	0.22
σ_{turb}	Year 1	0.05	0.16	0.21
	Year 2	0.03	0.02	0.22
σ_{FP}	Year 1	0.12	0.21	0.12
	Year 2	0.09	0.17	0.05
$\boldsymbol{\sigma_{FN2O}}$	**Year 1**	**0.18**	**0.41**	**0.32**
	Year 2	**0.17**	**0.32**	**0.32**

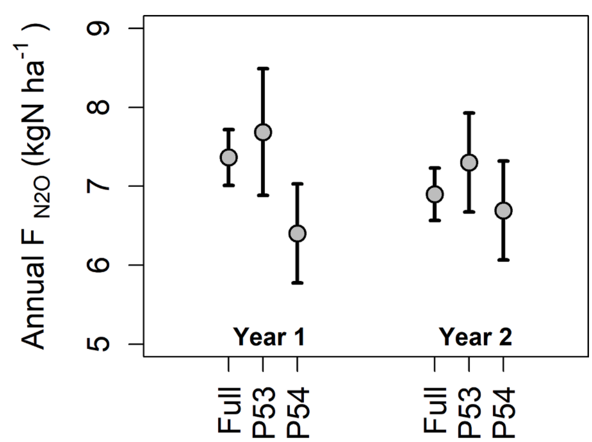

Fig. 6. Annual total F_{N2O} estimated using the kNN gap filling approach. Error bars are 95% confidence intervals.

algorithm.

Annual F_{N2O} for the full footprint, P53, and P54 were 7.4, 7.7, and 6.4 kg N_2O-N ha^{-1}, respectively in Year 1, and 6.9, 7.3, and 6.7 kg N_2O-N ha^{-1}, respectively in Year 2 (Fig. 6). As expected, the annual F_{N2O} from the full footprint falls between the separately partitioned paddocks, representing a mixture of the two source areas. Given that σ_{FN2O} effectively represents a standard deviation on the annual estimate, we calculated 95% confidence intervals (1.96* σ_{FN2O}) as the annual uncertainty, which ranged from 0.33-0.8 kg N_2O-N ha^{-1}. Given these 95% confidence intervals, the smallest annual difference in F_{N2O} between two adjacent paddocks that we can currently detect using one flux tower is conservatively 0.8 kg N_2O-N ha^{-1}, which was roughly 11% of annual emissions at this site.

4. Discussion

4.1. Improved gap filling approach

Overall the kNN algorithm accounted for the most variance and minimized bias relative to PLSR and NN. The approach presented here also generates gap-filling predictions that can recreate the structure of artificially withheld N_2O emission pulses that are wholly misrepresented using simple linear interpolation or running median approaches. Furthermore, despite relatively low percentage data coverage annually (< 12%), we could detect differences between two adjacent grazed paddocks as low as 11% (0.8 kg N_2O-N ha^{-1}), making this a viable approach for testing emissions mitigation options at field scales. The amount of information lost due to the footprint partitioning method could potentially be reduced by exploring other ways of treating spatial heterogeneity within an eddy covariance flux footprint (e.g., Levy et al., 2020), which would further improve adjacent paddock or other surface comparisons.

Gap filling missing F_{N2O} is particularly challenging because of the complexity of biogeochemical interactions responsible for the production of N_2O in the field across space and through time, for which we have incomplete understanding and information (Smith, 2017). This is particularly true of the pulse emissions that punctuate F_{N2O} time series in grazed systems (Liang et al., 2018) or other managed grasslands (e.g., Merbold et al., 2014; Cowan et al., 2020). While some statistical modelling approaches may generate better quality predictions than others (e.g. Taki et al., 2018), our analysis suggests that the creation of more comprehensive input features is critical. An advantage of machine learning techniques is that no *a priori* knowledge of the relationships between drivers and target variables is required to fit a model. However, offering as much information as possible in the form of additional features, even if only based on a general understanding of the system (e.g. days since grazing) can improve the chances of any algorithm recreating complex patterns. Our approach was to generate a large number of additional features based on the standard suite of meteorological and management data, including lagged (or cumulative) terms and cross products (Figure S2). Of course some of the information contained in these features is thus redundant, requiring additional treatment to avoid potential issues arising from multi-collinearity. A common approach in regression problems involves transforming the input matrix into orthogonal principal components, reducing the inputs to a subset that sufficiently represents the variance of the original input matrix. However, this approach does not discern any links to the target variable (i.e. F_{N2O}), and therefore the practitioner is again responsible for determining which input variables may be important, lest non-valuable information be included (or even over-represented) as input.

A recent analysis of gap filling approaches for eddy covariance methane fluxes tested the use of principle component inputs for several machine learning algorithms but ultimately found no improvement over the non-transformed inputs (Kim et al., 2020). Here, we have presented a more valuable way to treat inputs for machine learning gap filling approaches that optimizes the available information by way of PLS transformation, without requiring the process understanding that would ultimately be preferable. This approach may also have improved the performance of machine learning algorithms tested by Kim et al. (2020)), and may have applicability for CO_2 flux measurements as well. The kNN regression using PLS inputs generated for F_{N2O} here, resulted in more variance explained in the validation data and less bias across the distribution than using principle components calculated from the same input matrices (not shown). It is also worth noting that we examined results obtained when omitting the PLS dimension reduction step (i.e. using the full set of inputs and features for each footprint). This slightly improved the variance explained in validation data and reduced bias. The minor improvement may have resulted from the extra information contained in the raw input matrix, but the strong correlation of many raw input variables also increases the risk of diminishing generalizability between the training and validation datasets. Additionally, because of the large size of the full input matrix, processing time increased more than ten-fold, which would render this approach unsuitable to most practitioners. The resulting annual numbers were also not meaningfully different, adding confidence to the more robust PLS input approach.

While the NN and kNN algorithms produced output of generally similar quality, kNN metrics suggested a slightly better performance here, particularly in reducing bias at the high end of the distribution. We also explored the use of general additive models (GAM) and random forests, finding similar, but not better results from the validation data set metrics (not shown). The kNN gap-filled F_{N2O} performed similarly well across all three footprint designations (full, P53, and P54) despite a higher percentage of large gaps in the footprint-partitioned data. The primary shortcoming of the approach we have presented is the remaining under-estimation of large magnitude fluxes. However, the contribution of fluxes greater than 4 nmol m^{-2} s^{-1}, where this bias is greatest, to the annual total F_{N2O} was less than 8% on average across the three source areas. Therefore, under-estimating this part of the distribution is of lesser concern.

The estimates from linear interpolation consistently resulted in 0.5-1.0 kg N_2O-N ha^{-1} larger annual sums than kNN, while 30-day median gap filling resulted in 0.6-1.0 kg N_2O-N ha^{-1} lower annual F_{N2O}. In addition to providing examples to illustrate how using either of these approaches can be problematic for situations with long gaps, methods for estimating uncertainty associated with these are not available and we cannot therefore compare estimates among paddocks or years with confidence. Furthermore, depending on the integral time period of interest (e.g., season, year etc.), the quality of the gap-filling approach can become more important and the necessity of a realistic uncertainty estimate may be enhanced for shorter period comparisons. Both Kroon et al. (2010) and Nemitz et al. (2018) have suggested detailed methodologies for estimating random uncertainties associated with individual half-hourly F_{N2O}. While others have developed Bayesian approaches that offer a robust strategy, intrinsically providing uncertainties in both model and measurements (Levy et al., 2017), prior model structure and parameter distributions must be defined, which may be complicated in grazed systems. Our approach to uncertainties was intended as a pragmatic way to estimate all potential sources contained within a given measurement or gap-filled value using accepted techniques within the CO_2 flux literature. For example, calculating the differences between adjacent half-hourly measurements is easy to execute and results in the same heteroscedastic behaviour (Figure S4) expected with any scalar flux, and also demonstrated by Kroon et al. (2010) for F_{N2O}.

4.2. Implications

Using a single flux tower and a relatively comprehensive estimate of uncertainty, we were able to resolve apparent differences in annual F_{N2O} of ≤ 11% over adjacent paddocks in a grazed dairy farm. Small-scale studies of mitigation strategies for reducing N_2O emissions from managed grasslands typically find > 30% lower emissions using biological (different pasture species) or chemical inhibitors (dicyandiamide) (e.g., Akiyam et al., 2010; Smith et al., 2008b; Lou et al., 2018). To date, however, it has been difficult to evaluate mitigation attempts at field-scale with traditional techniques because of the spatiotemporal heterogeneity of F_{N2O} and the limitations of chambers and scaling approaches to capture that variability (e.g., Wecking et al., 2020). For example, assessment at field-scale is critical because spatial scaling using chamber data cannot easily account for aligned farming practices such as consequences of animal compaction of soil with urine or dung patches at different moisture contents. Here we found that despite renewing the pasture in one paddock (P54) to include plantain in an effort to mitigate F_{N2O}, there was not an attributable reduction in the annual total. This may have been the result of higher fluxes during the renewal period itself, during which the new pasture species were still establishing, and insufficient time to detect the expected mitigation effects, but this is the subject of ongoing work with a longer record.

It is still uncommon for researchers to obtain multiple eddy covariance-capable N_2O analysers to deploy towers in control and treatment plots due to the expense. Because of this, it is necessary to test the validity of using one flux tower situated at a boundary between parcels with different management practices (Fuchs et al., 2018). In grazed systems where pulse emissions are more variable through time than harvested grasslands or crops, the gap filling strategy becomes a key factor in determining the success of the approach. Our approach suggests that we can realistically discern mitigation-induced differences in annual F_{N2O} that are relevant to emissions reduction strategies. Furthermore, this can be coupled with CO_2 and CH_4 flux observations to obtain full GHG balances for adjacent paddocks, which is essential in determining the relative effects of any given mitigation strategy.

5. Conclusions

Our aim was to develop a robust gap filling approach for eddy covariance F_{N2O} and estimate uncertainties to determine the minimum detectable difference in annual totals between two adjacent grazed pastures. Given the complexity of processes driving F_{N2O}, we generated a large set of input features rather than limit inputs based on perceived importance. These inputs were transformed using PLS decomposition to avoid offering redundant information and reduce processing time, and fed into machine learning algorithms. We found that kNN regression generated the best predictions of missing F_{N2O} based on validation data, particularly during pulse emissions, and we used these to fill gaps in F_{N2O} partitioned by footprint contribution into two adjacent grazed paddocks on a commercially operating dairy farm in New Zealand. Annual uncertainties were estimated by combining components from random measurements, gap filling accuracy and precision, and threshold choices for atmospheric turbulence and footprint contribution. Total N_2O emissions for the full footprint, P53, and P54 were 7.4 $\pm$0.35, 7.7 $\pm$0.80, and 6.4 $\pm$0.63 kg N_2O-N ha^{-1}, respectively in Year 1, and 6.9 $\pm$0.33, 7.3 $\pm$0.63, and 6.7 $\pm$0.63 kg N_2O-N ha^{-1}, respectively in Year 2. These 95% confidence intervals on the annual F_{N2O} suggest that we could detect differences on the order of 10-15% at this site, which is a meaningful degree, considering the expected (or intended) effect of available mitigation strategies for reducing agricultural emissions.

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Competing Interest

None.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.agrformet.2020.108280.

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